

Workload Metrics computation using GPS & Heart Rate data

I – Basic Movement metrics:

1- Total Distance (meters/kilometers):

- Use Haversine formula to compute distance between two consecutive points
- Latitude & Longitude are used to compute the distance between two points
- Total distance = sum of all dx

2- Speed / Pace – Instantaneous, Average & Max (m/s, km/h, min/km):

- Pace is the inverse of speed: $\text{Pace} = \text{time}/\text{distance}$ | $\text{Speed} = \text{distance}/\text{time}$
- Instantaneous Pace = dt/dx (between two recorded points)
- Average pace = overall time / overall distance
- Max pace is the highest pace recorded during the whole activity

3- Moving Time / Stopped Time (time):

- Average human walking speed is about 1.4 m/s
- Stopped time = sum of time when instantaneous speed is lower than a defined threshold
- Moving time = overall time – stopped time
- Strava uses threshold = 0.9 m/s (=stopped)

4- Acceleration / Deceleration (rate of speed change):

- $\text{Acceleration} = (\text{speed_final} - \text{speed_initial}) / \text{time}$
- if positive (acceleration), if negative (deceleration)
- Use “instantaneous acceleration impulse” ? But no sign (compare the two speed points)

5- Route / Path (visualisation of trajectory on map):

- WORK IN PROGRESS... FOR LATER

II – Segmented Performance metrics:

1- Split Pace (average pace per km):

- Use instantaneous pace computed previously and divide total distance by km, then do the average pace in min/km

2- Best segment times (fastest 1km, 5km, 10km):

- Depending on how long was the activity, divide into different segments (distance).
- Compare average pace over each segment and determine shortest time

3- Time in speed zones (percentage of time spent: walking, jogging, running, sprinting):

- Determine walking (0-7km/h), jogging (7-14km/h), running (14-19.8km/h) and sprinting (>19.8km/h) speeds threshold, high-intense sprint (>25km/h)
- Create a list for each of the speed zones and sum-up the times
- Get the percentage of every zone by comparing with overall time

III – Cardiovascular Load metrics:

1- Average HR (in BPM):

- Do a simple average of the overall HR data in bpm.

2- Max HR (peak HR reached):

- Iterate through the HR data and find the highest value
- Do a for loop through the HR data and “keep in mind” the biggest value yet.

3- Resting HR (useful for calculating intensity) **[IF BASELINE AVAILABLE]:**

- the resting HR is to be set by the user (specific to each player) and cannot be computed

4- HR Reserve HRR (= $H_{\text{rmax}} - H_{\text{Rrest}}$):

- If the resting HR is provided, then the formula is simple: $\text{HRR} = \text{max} - \text{rest}$

5- Time in HR zones (time spent in %HRmax zones eg: Z1: 50-60%, Z5: 90-100%):

- HRmax varies for each person ($= 220 - \text{age}$) [OR USE MAX HR REACHED DURING EXERCISE], so a good metric is evaluated with % of HRmax.
- Here we want to compute the time spent in each HR zone for training
- Z1: 50-60% | Z2: 60-70% | Z3: 70-80% | Z4: 80-90% | Z5: 90-100%
- HRmax for the user should be provided in the config parameters
- Compute intervals for each zone, and do a for loop through the HR data to sum up time for each HR zone

6- Training Impulse TRIMP (weighted measure of cardiovascular load):

- $\text{TRIMP} = \text{overall time (mins)} \times \text{HRR} \times \text{weight factor}$
- in this formula, $\text{HRR} = (\text{Avg HR} - \text{HR}_{\text{rest}}) / (\text{HR}_{\text{max}} - \text{HR}_{\text{rest}})$
- $\text{weight factor} = 0.64 \times e^{(1.67 \times \text{HRR})}$

7- HR-based intensity distribution (proportion of time in low, moderate, high effort):

- Same as point 5... but maybe showing it in a graph/pie chart for easy understanding.

IV – Efficiency & Fitness Indicators:

1- Pace vs HR [HR drift/decoupling] (indicates aerobic efficiency):

- $\text{Efficiency Factor} = \text{Avg Pace} / \text{Avg HR}$
- this factor indicates how much speed you generate per heartbeat.
- Higher EF → greater aerobic efficiency (running same pace at lower heart rate)

2- Running Economy (pace per heartbeat):

- $\text{Efficiency Factor} = \text{Avg Pace} / \text{Avg HR}$
- Then, a higher Efficiency Factor (EF) → greater running efficiency (body is covering more distance per heartbeat)

3- HR Rate Recovery HR_{rec} (rate of HR drop post-exercise):

- $\text{HR recovery} = \text{HR}_{\text{max}} - \text{HR}_{1\text{min_after_workout}}$
- Higher value indicates better cardiovascular fitness (recovered a lot in 1 minute...)
- DATA IS NOT AVAILABLE IN OUR CASE...

4- Cardiac Cost (HR per unit distance) [IF AVAILABLE]:

- Take the Avg HR (in bpm) and determine the total number of heartbeat during the activity:
 $\text{Heartbeats} = \text{Avg HR} \times \text{overall time (mins)}$
- $\text{Cardiac Cost (beats per Km)} = \text{Heartbeats} / \text{total distance (km)}$
- lower cardiac cost per unit distance → greater efficiency (heart beats fewer times to cover the same distance)